

State Machine Question (100 marks)

Given the problem description below, specify the *state machine* for the watch. **Show the states as well as transitions including events, guards, and actions. Define composite states and orthogonal regions when needed to reduce complexity.** You do not need to specify parameters for events, guards, and actions but names of actions must clearly show when variables are initialized (and to which value) or updated (and to which value) as well as all changes to the watch display. You may assume that changing a variable will automatically update the watch display. Use meaningful names for the private methods for guards and actions in your state machine. You do not need to define the method bodies of those methods.

Problem Description: The watch W-UMPLE combines time keeping with alarm and countdown functionalities. The watch shows the time in hours, minutes, and seconds. The watch has four buttons (A, B, C, and D). Press A to change from the timekeeping mode to the alarm mode. If A is pressed another time, then the watch is in the countdown time mode. If A is pressed one more time, then the watch is back in the timekeeping mode.

Press D to turn on the backlight to illuminate the watch display. The backlight can be turned off by pressing D again and will automatically turn off after 10 seconds.

In the timekeeping mode, the current time is shown. Press C to set the current time. First, the seconds can be set. Press C to rotate from seconds to minutes to hours and back to seconds. When seconds, minutes, or hours can be set, they will flash. Press B to change seconds, minutes, or hours in increments of one. Press A to save the changes, i.e., the seconds, minutes, or hours do not flash anymore and cannot be set and the current time is shown.

In the alarm mode, the current alarm time is shown. Press C to set the alarm time. First, the minutes can be set. Press C to rotate from minutes to hours and back to minutes. When minutes or hours can be set, they will flash. Press B to change minutes or hours in increments of one. Press A to save the changes and return to the timekeeping mode. In the timekeeping mode, press B to turn the alarm on and off. Press B for at least 2 seconds to turn Flash Alert on and off. If Flash Alert is turned on, the backlight will flash for 5 seconds in addition to the alarm time flashing for 5 seconds and the watch beeping for 5 seconds when the alarm goes off. The alarm notifications can be stopped by pressing any button. After the alarm, the watch returns to where it was before the alarm went off.

In the countdown mode, the current countdown time is shown. Press C to set the countdown time. First, the seconds can be set. Press C to rotate from seconds to minutes to hours and back to seconds. When seconds, minutes, or hours can be set, they will flash. Press B to change seconds, minutes, or hours in increments of one. Press A to save the changes and return to the countdown mode. In the countdown mode, press B to start the countdown (if a countdown time is set) and B again to stop the countdown. When the end of the countdown is reached, 0:00:00 will flash for 5 seconds and the watch will beep for 5 seconds. If Flash Alert is turned on, the backlight will also flash for 5 seconds. The countdown notifications can be stopped by pressing any button. After the countdown notifications, the watch is in the countdown mode.